

# Lecture04: Loop, Number, math and String

## if-else in Js

It's like deciding what to wear:

1. **Question 1 (if):** "Is it raining?"
    - **YES?** -> Put on a raincoat. (Stop here, you're done.)
    - **NO?** -> Move to the next question.
  2. **Question 2 (else if):** "Okay, is it sunny?"
    - **YES?** -> Put on sunglasses. (Stop here, you're done.)
    - **NO?** -> Move to the next question.
  3. **The Fallback (else):** "Since nothing else was true..."
    - Just wear a regular sweater.
- 

## 1. The Basic if Statement

The if statement is the starting point. It checks a single condition. If that condition is true, the code inside its curly braces {} will run. If the condition is false, the code block is completely skipped.

**Structure:**

codeJavaScript

```
if (condition) { // This code runs only if the condition is true. }
```

**Example:**

codeJavaScript

```
let temperature = 30; if (temperature > 25) { console.log("It's a hot day! Wear shorts."); } // Code continues here...
```

**Output:**

```
It's a hot day! Wear shorts.
```

If we set temperature to 15, the condition  $15 > 25$  would be false, and nothing would be printed.

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## 2. The if...else Statement

The else statement provides a "fallback" or "alternative" action. It runs only when the initial if condition is false.

**Structure:**

```
if (condition) { // This code runs if the condition is true. } else { // This code runs
if the condition is false. }
```

**Example:**

```
let age = 16; if (age >= 18) { console.log("You are old enough to vote."); } else {
console.log("You are not old enough to vote yet."); }
```

**Output:**

You are not old enough to vote yet.

One of these two blocks is **guaranteed** to run.

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### 3. The if...else if...else Chain

This is the full decision-making chain. It allows you to check multiple, different conditions in order.

**Key Rule:** The chain is evaluated from top to bottom. The very **first condition that is true** gets its code block executed, and the rest of the entire chain is skipped.

**Structure:**

```
if (condition1) { // Runs if condition1 is true. } else if (condition2) { // Runs if  
condition1 is false AND condition2 is true. } else if (condition3) { // Runs if 1 and 2  
are false AND condition3 is true. } else { // Runs if ALL previous conditions were  
false. }
```

**Example: Grading a test score**

This is a classic use case. The order is very important here.

```
let score = 85; let grade; if (score >= 90) { grade = 'A'; } else if (score >= 80) { grade = 'B'; } else if (score >= 70) { grade = 'C'; } else if (score >= 60) { grade = 'D'; } else { grade = 'F'; } console.log(`Your grade is: ${grade}`);
```

## Loop in JS

### 1. The for Loop

The for loop is the most common type of loop. It's perfect when you know **exactly how many times you want to repeat** an action.

Think of it like setting a timer for a specific number of repetitions.

**Structure:**

The for loop has a specific structure with three parts inside its parentheses, separated by semicolons:

for (initialization; condition; final-expression) { ... }

1. **Initialization:** This runs **only once** at the very beginning. It's where you create your counter variable (traditionally named *i* for "index").
2. **Condition:** This is checked **before each repetition**. If the condition is true, the code inside the loop runs. If it becomes false, the loop stops.
3. **Final-Expression:** This runs **after each repetition**. It's where you usually increment your counter.

```
// Initialization: let i = 1; (Start a counter at 1) // Condition: i <= 5; (Keep looping  
as long as i is less than or equal to 5) // Final-Expression: i++; (After each loop, add  
1 to i) for (let i = 1; i <= 5; i++) { console.log("This is repetition number:", i); }
```

Output:

```
This is repetition number: 1 This is repetition number: 2 This is repetition number: 3  
This is repetition number: 4 This is repetition number: 5
```

The loop stops because after  $i$  becomes 5 and the code runs, the  $i++$  makes  $i$  become 6. The condition  $6 \leq 5$  is now false, so the loop terminates.

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## 2. The while Loop

The while loop is simpler. It's perfect when you want to **keep looping as long as a certain condition is true**, but you don't necessarily know ahead of time how many repetitions that will be.

Think of it as saying, "Keep doing this *while* this is true."

### Structure:

```
while (condition) { ... }
```

The while loop only has a condition. The initialization must happen *before* the loop, and the final-expression (the update to the variable) must happen *inside* the loop.

**Example:** A simple game loop that runs until the player runs out of health.

```
let playerHealth = 10; while (playerHealth > 0) { console.log(`Player health is  
${playerHealth}. Attacking monster!`); // Inside the loop, we must change the variable  
to avoid an infinite loop playerHealth -= 3; // Player takes 3 damage if (playerHealth  
<= 0) { console.log("Player has been defeated!"); } }
```

Output:

```
Player health is 10. Attacking monster! Player health is 7. Attacking monster! Player  
health is 4. Attacking monster! Player health is 1. Attacking monster! Player has been  
defeated!
```

**Critical Danger:** If you forget to change the condition variable inside a while loop (like forgetting `playerHealth -= 3;`), the condition will always be true, and your program will get stuck in an **infinite loop** and crash.

### 3. The do...while Loop

This is a less common variation of the while loop. Its unique feature is that the code inside the loop is **guaranteed to run at least once**.



The condition is checked **after** the code block runs, not before.

### Structure:

```
do { ... } while (condition);
```

Think of it as "Do this, and then check if you should do it again."

**Example: Asking the user for input until they provide a valid response.**

You always want to ask the user at least once.

```
let userResponse; do { // This code will always run at least one time userResponse =  
  prompt("Please type 'yes' to continue:"); // The prompt() function shows a popup in the  
  browser } while (userResponse !== "yes"); console.log("You typed 'yes'. Thank you!");
```

In this example, the prompt will keep appearing until the user types the exact word "yes". The first prompt is guaranteed to show.

## Number

### 1. Introduction: The Unified Number Type

In JavaScript, there is only one type for numbers: `number`. This single type is used to represent both integers (whole numbers like `10`, `-50`) and floating-point numbers (decimals like `3.14`, `-0.5`).

**The Core Standard:** All numbers in JavaScript are implemented as **64-bit double-precision floating-point numbers**, following the international **IEEE 754 standard**. This is the same format used for `double` in languages like C++ and Java.

```
let integer = 100; let float = 99.5; console.log(typeof integer); // "number" console.log(typeof float); // "number"
```

## 2. Creating Numbers

You can create numbers in several ways:

- **Standard Literals:**

```
let a = 25; // Integer let b = 12.34; // Floating-point
```

- **Exponential Notation (e):** A shorthand for writing very large or very small numbers.

```
let billion = 1e9; // 1 followed by 9 zeros -> 1000000000
let tiny = 5e-6; // 5 / 10^6 -> 0.000005
```

- **Other Bases (Hex, Binary, Octal):** You can also represent numbers in other numeral systems.

```
let hex = 0xFF; // Hexadecimal (base 16) -> 255 in decimal
let binary = 0b1010; // Binary (base 2) -> 10 in decimal
let octal = 0o77; // Octal (base 8) -> 63 in decimal
```

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### 3. The "Gotcha": Floating-Point Inaccuracy

Because numbers are stored in a binary floating-point format (base-2), they cannot perfectly represent all decimal fractions (base-10). This leads to the most famous "gotcha" in JavaScript math.

```
console.log(0.1 + 0.2); // Outputs: 0.30000000000000004
console.log(0.1 + 0.2 === 0.3); // false
```

**Why?** The numbers `0.1` and `0.2` (and `0.3`) have infinitely repeating representations in binary, similar to how `1/3` is `0.333...` in decimal. The computer has to round them off to fit them into the 64-bit storage. These tiny rounding errors accumulate.

**How to handle this:**

1. **For financial calculations:** Never use floating-point numbers. Work with integers (e.g., store money in cents).
  2. **For display:** Use the `.toFixed()` method to round the result to a specific number of decimal places.
  3. **For comparison:** Check if two numbers are "close enough" using `Number.EPSILON`.
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## 4. Special Numeric Values

There are three special values that are technically of type `number`.

- **Infinity**: Represents a value larger than the largest possible number. It results from division by zero or exceeding `Number.MAX_VALUE`.

```
console.log(1 / 0); // Infinity console.log(-1 / 0); // -Infinity console.log(typeof Infinity); // "number"
```

- **Infinity**: Represents a value smaller than the smallest possible number.

- **NaN (Not a Number)**: Represents the result of an invalid or undefined mathematical operation. It's the error code for failed math. **Key Property of NaN**: It is the only value in JavaScript that is not equal to itself.

```
console.log("hello" / 2); // NaN console.log(Math.sqrt(-1)); // NaN console.log(typeof NaN); // "number"
```

```
console.log(NaN === NaN); // false
```

## 5. Important **Number** Properties and Methods

The **Number** object provides several useful constants and methods.

### A. **Number** Constants (Limits of the Type)

- **Number.MAX\_VALUE**: The largest positive number that can be represented ( $\sim 1.8e+308$ ).
- **Number.MIN\_VALUE**: The smallest positive number closest to zero.
- **Number.MAX\_SAFE\_INTEGER**: The largest integer that can be safely represented without losing precision ( $2^{53} - 1$ ). **This is the one you should care about for integer math.**
- **Number.MIN\_SAFE\_INTEGER**: The smallest safe integer.

- **Number.EPSILON**: The smallest interval between two representable numbers. Used for floating-point comparisons.

## B. Checking Number Types

- **isNaN(value)** (Global function): The classic way to check if a value is **NaN**. **Gotcha**: It coerces the value first, so **isNaN("hello")** is **true**, which can be misleading.
- **Number.isNaN(value)** (Modern - ES6): The better, more reliable way. It returns **true** only if the value is actually **NaN**. It does not perform type coercion.

```
isNaN("blue"); // true (coerces "blue" to NaN) Number.isNaN("blue"); // false (it's  
a string, not NaN) let result = 0 / 0; // result is NaN isNaN(result); // true Num  
ber.isNaN(result); // true
```

- **isFinite(value)** / **Number.isFinite(value)**: Checks if a value is a real, finite number (not **Infinity**, **Infinity**, or **NaN**). The **Number.isFinite()** version is stricter and does not coerce.
- **Number.isInteger(value)**: Checks if a value is an integer.

## C. Formatting and Converting Numbers

These methods are called on a number variable.

- **.toString(base)** : Converts a number to a string. You can optionally provide a base (from 2 to 36).

```
let num = 255; console.log(num.toString()); // "255" (base 10 - default) console.log(num.toString(16)); // "ff" (base 16 - hexadecimal) console.log(num.toString(2)); // "11111111" (base 2 - binary)
```

- **.toFixed(digits)** : Formats a number to a fixed number of decimal places and returns a **string**. Useful for formatting currency.

```
let price = 19.991234; console.log(price.toFixed(2)); // "19.99"
```

- **.toPrecision(digits)** : Formats a number to a specified total number of significant digits and returns a **string**.

```
let n = 123.456; console.log(n.toPrecision(4)); // "123.5" (4 significant digits)
```

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## 6. The **Math** Object

JavaScript provides a built-in `Math` object that has properties and methods for mathematical constants and functions. It is not a constructor; you use it directly.

- **Constants:** `Math.PI`, `Math.E`
- **Rounding:**
  - `Math.round(x)` : Standard rounding to the nearest integer.
  - `Math.floor(x)` : Rounds **down** to the nearest integer.
  - `Math.ceil(x)` : Rounds **up** to the nearest integer.
  - `Math.trunc(x)` : Removes the decimal part, leaving only the integer (truncates toward zero).
- **Other Common Functions:**
  - `Math.abs(x)` : Absolute value.
  - `Math.pow(x, y)` : `x` to the power of `y` (same as `x ** y`).
  - `Math.sqrt(x)` : Square root.
  - `Math.max(a, b, c...)` : Returns the largest of the given numbers.
  - `Math.min(a, b, c...)` : Returns the smallest.
  - `Math.random()` : Returns a pseudo-random number between 0 (inclusive) and 1 (exclusive).

```
console.log(Math.round(4.7)); // 5 console.log(Math.floor(4.7)); // 4 console.log(Math.ceil(4.2)); // 5 console.log(Math.max(10, -5, 100, 0)); // 100 // To get a random integer between 1 and 10 (inclusive): let randomInt = Math.floor(Math.random() * 10) + 1;
```



## The Core Tool: `Math.random()`

JavaScript's only built-in tool for randomness is `Math.random()`. You must understand its behavior perfectly to build anything else.

`Math.random()` returns a random floating-point number between `0` (inclusive) and `1` (exclusive).

This means it can return `0`, `0.1234`, `0.5`, `0.99999`, but it will **never** return `1.0`.

- Range: `[0, 1)`
- 

## The Goal: A Random Integer Between `min` and `max` (Inclusive)

Let's say our goal is to get a random whole number from 1 to 10. We want `1`, `2`, `3`, ..., `9`, `10`.

The formula to achieve this is:

```
Math.floor(Math.random() * (max - min + 1)) + min
```

This looks complicated, so let's build it up step-by-step to understand the intuition.

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## Building the Formula from First Principles

Let's stick with our goal: a random integer between **1 (min)** and **10 (max)**.

### Step 1: Scale up the range.

`Math.random()` gives us a number between `0` and `1` (e.g., `0.5`). We want a number in a much larger range. How many different numbers do we want to be able to generate?

`1, 2, 3, 4, 5, 6, 7, 8, 9, 10`

There are **10** possible outcomes.

The number of outcomes is always `max - min + 1`.

`10 - 1 + 1 = 10`.

So, the first step is to scale our `[0, 1)` range up to a range of the correct size. We do this by multiplying.

```
Math.random() * 10;
```

- The lowest possible value is `0 * 10 = 0`.
- The highest possible value is `0.999... * 10 = 9.999...`.
- Our new range is `[0, 10)`.

This gives us a floating-point number from `0` up to (but not including) `10`.

## Step 2: Get rid of the decimals.

We want whole numbers (integers). The perfect tool for this is `Math.floor()`, which always rounds **down** to the nearest whole number.

Let's see what happens when we apply `Math.floor()` to our range `[0, 10)`:

```
Math.floor(Math.random() * 10);
```

- If `Math.random()` was `0.0`, `Math.floor(0)` is `0`.
- If `Math.random()` was `0.123`, `Math.floor(1.23)` is `1`.
- If `Math.random()` was `0.999`, `Math.floor(9.99)` is `9`.

Applying `Math.floor()` has transformed our float range `[0, 10)` into an integer range of `0, 1, 2, 3, 4, 5, 6, 7, 8, 9`.

This is close! We have 10 possible numbers, but the range is wrong. We want `1` to `10`, not `0` to `9`.

## Step 3: Shift the range to the correct starting point.

Our current range is `[0, 9]`. We want `[1, 10]`.

How do you get from 0 to 1? You add 1.

How do you get from 9 to 10? You add 1.

The final step is to shift our entire range up by adding our desired minimum value (`min`).

```
Math.floor(Math.random() * 10) + 1;
```

Let's trace the outcomes:

- If the `Math.floor()` part gave us `0`, the result is `0 + 1 = 1`.
- If the `Math.floor()` part gave us `9`, the result is `9 + 1 = 10`.

This formula now correctly produces a random integer from 1 to 10, inclusive.

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## The General-Purpose Function

By replacing the specific numbers `10` and `1` with `max` and `min`, we get our general-purpose formula.

```
/** * Generates a random integer between a minimum and maximum value (inclusive). * @param {number} min The minimum possible value. * @param {number} max The maximum possible value. * @returns {number} A random integer within the range. */ function getRandomInt(min, max) { // 1. Calculate the number of possible outcomes (the size of our range). const range = max - min + 1; // 2. Scale up Math.random() to create a float in the range [0, range). const scaled = Math.random() * range; // 3. Round down to get an integer in the range [0, range-1]. const floored = Math.floor(scaled); // 4. Shift the range up to [min, max] by adding the minimum value. const result = floored + min; return result; } // Example usage: console.log("Random number between 1 and 10:", getRandomInt(1, 10)); console.log("Random number between 50 and 100:", getRandomInt(50, 100)); console.log("Random dice roll (1 to 6):", getRandomInt(1, 6));
```

## 1. Introduction: What is a String?

A string is a primitive data type in JavaScript used to represent a sequence of characters. Anything you can type—letters, numbers, symbols, punctuation—can be part of a string. It is the primary way we work with textual data.

### Key Characteristics:

- **It's a primitive:** This means strings are **immutable**.
  - **It's indexed:** Each character in a string has a numerical position (index), starting from zero.
  - **It's an object-like primitive:** Although it's a primitive, it has methods and properties we can use, like `.length`. JavaScript temporarily wraps it in a String object when you try to access these.
- 

## 2. Creating Strings

There are three ways to create a string literal in JavaScript.

### 1. Single Quotes ( `'...'` ):

```
let singleQuoted = 'Hello, world!';
```

### 2. Double Quotes ( `"..."` ): Functionally identical to single quotes. The main reason to choose one over the other is for convenience when a string itself contains quotes.

```
let doubleQuoted = "He said, 'Hello!'"; // Easy to include single quotes  
let singleQuotedWithDouble = 'She replied, "Hi!"'; // Easy to include double quotes
```

3. **Template Literals** (``...`` - ES6): The most powerful and modern way. They use backticks. We will cover the special features of template literals later.

```
let templateLiteral = `This is a template literal.`;
```

---

### 3. Core Properties and Concepts

#### A. The `.length` Property

Every string has a `.length` property that tells you how many characters it contains.

```
let greeting = "Hello"; console.log(greeting.length); // Outputs: 5
let emptyString = ""; console.log(emptyString.length); // Outputs: 0
```

#### B. Accessing Individual Characters (Zero-Based Indexing)

You can access a character at a specific position using square bracket notation `[]`. The first character is at index `0`.

```
let message = "JavaScript"; // J a v a S c r i p t // 0 1 2 3 4 5 6 7 8 9 console.log(message[0]); // "J" console.log(message[4]); // "S" // To get the last character, a common pattern is used: console.log(message[message.length - 1]); // "t"
```

### C. The Golden Rule: Strings are Immutable

This is the most critical concept. **You cannot change a string in place.** Any method that appears to modify a string will always **return a brand new string**, leaving the original untouched.

```
let name = "alex"; // Let's try to change the first character. name[0] = "A"; // This will FAIL silently. It does nothing. console.log(name); // Outputs: "alex" (The original string is unchanged) // Let's use a method that "changes" the string. let upperName = name.toUpperCase(); console.log(upperName); // Outputs: "ALEX" (This is a NEW string) console.log(name); // Outputs: "alex" (The original is still unchanged)
```

## 4. Common and Essential String Methods

These methods are your primary tools for working with strings. Remember, they all return **new strings**.

### A. Changing Case

- `.toUpperCase()` : Returns a new string with all characters in uppercase.



- `.toLowerCase()` : Returns a new string with all characters in lowercase.

```
let whisper = "please be quiet"; let shout = whisper.toUpperCase(); // "PLEASE BE QUIET"
```

## B. Finding Substrings

- `.indexOf(substring)` : Returns the index of the **first occurrence** of a substring. If the substring is not found, it returns -1.
- `.lastIndexOf(substring)` : Returns the index of the **last occurrence** of a substring. Returns -1 if not found.
- `.includes(substring)` (ES6): Returns a boolean (`true` or `false`) indicating if the string contains the substring. This is often more readable than `indexOf`.

```
let sentence = "The quick brown fox jumps over the lazy fox."; console.log(sentence.indexOf("fox")); // 16 (the first one) console.log(sentence.lastIndexOf("fox")); // 40 (the last one) console.log(sentence.indexOf("cat")); // -1 (not found) console.log(sentence.includes("jumps")); // true console.log(sentence.includes("cat")); // false
```

## C. Extracting Substrings

- `.slice(startIndex, endIndex)` : Extracts a section of a string and returns it as a new string.
  - `startIndex` : The index where the extraction begins (inclusive).
  - `endIndex` : The index where the extraction ends (exclusive - it does not include this character).
  - You can use negative indices, which count from the end of the string.

```
let text = "JavaScript"; console.log(text.slice(0, 4)); // "Java" (from index 0 up to, but not including, 4) console.log(text.slice(4)); // "Script" (from index 4 to the end) console.log(text.slice(-6)); // "Script" (the last 6 characters)
```

- `.substring(startIndex, endIndex)` : Similar to `.slice()` , but it doesn't accept negative indices.
- `.substr(startIndex, length)` : (Deprecated - Avoid). It works with a start index and a length, which is different from the others. Use `.slice()` instead.

#### D. Replacing Substrings

- `.replace(searchValue, newValue)` : Finds a `searchValue` and replaces it with `newValue` .
  - **Gotcha:** By default, it only replaces the **first occurrence**.
- `.replaceAll(searchValue, newValue)` (ES2021): Replaces **all occurrences**. This is the modern, easy way to do a global replace.

```
let greeting = "hello world, hello there"; // Replaces only the first "hello"
let newGreeting = greeting.replace("hello", "hi"); console.log(newGreeting); // "hi world, hello t
here" // Replaces all "hello"s
let allNewGreeting = greeting.replaceAll("hello", "hi");
console.log(allNewGreeting); // "hi world, hi there"
```

- **Old way to replace all:** Use a regular expression with the global flag ( `/g` ).

```
greeting.replace(/hello/g, "hi");
```

## E. Cleaning Up Whitespace

- `.trim()` : Removes whitespace from both the beginning and end of a string.
- `.trimStart()` / `.trimEnd()` : Removes whitespace from only the start or end.

```
let userInput = " my-username@example.com "; let cleanedInput = userInput.trim();
// "my-username@example.com"
```

## F. Splitting a String into an Array

- **.split(separator)** : Splits a string into an array of substrings, using the **separator** to decide where to split. This is incredibly useful.

```
let csvData = "item1,item2,item3"; let items = csvData.split(","); // ["item1", "item2", "item3"]
let words = "The quick brown fox"; let word_array = words.split(" "); // ["The", "quick", "brown", "fox"]
let letters = "abc"; let letter_array = letters.split(""); // ["a", "b", "c"]
```

---

## 5. Template Literals (ES6) - The Modern Way

Template literals, created with backticks ( ``` ), are a massive improvement for working with strings.

### A. Variable Interpolation ( `${...}` ):

This allows you to embed expressions and variables directly into a string. It's far more readable than using the `+` operator for concatenation.

- The Old Way (Concatenation):

```
let name = "Alice"; let age = 30; let message = "Hello, my name is " + name + " and I am " + age + " years old.";
```

- The New Way (Template Literals):